# Sama Al-Oda

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## Education

**McMaster University** Expected May 2028
Bachelor of Software Engineering, Double Major with Biomedical Engineering – GPA: 3.96

## Experience

**Software Engineering Intern (CyberDome / R&D)** May 2026 – Aug 2026
Nokia | *C, Python, Kubernetes, Docker, Helm, Jenkins, Ansible* Ottawa, ON

- Root-caused a production alert flood to bad live telemetry, then fixed the C collector so false alarms dropped from **580+/s to ∼1/s** with **no lost events** – turning an unusable detector back into a reliable signal.
- Debugged a silent telemetry failure where services appeared healthy but logs never reached monitoring, then validated end-to-end delivery across **four deployment setups** to catch failures before release.
- Closed the loop from **commit** → **image** → **install bundle**, automating branch packaging and ensuring offline installs.
- Designed a backup and recovery pipeline around a storage limit that blocked the usual approach, then proved **backup** → **wipe** → **restore** across deployments so learned system state could be recovered reliably instead of rebuilt from scratch.

**Software Developer Intern** May 2025 – Apr 2026
Didar Lab / FendX Technologies | *Python, Swift, AWS, PostgreSQL, REST APIs* Hamilton, ON

- Shipped **AWS-backed REST APIs** for concurrent mobile clients, covering authentication, session handling, and persistence while maintaining typical request latency under **200ms**.
- Built and maintained **iOS and Android** clients in **Swift** and Gradle, integrating **PostgreSQL, Supabase, and Firebase** to keep multi-user workflows consistent across devices and across the full client-to-backend data flow.
- Turned research prototypes into product-ready services by wrapping inference behind APIs, automating data-processing pipelines, and hardening the path from notebook experiments to deployed software used in **real-world mobile workflows**.

## Projects & Hackathons

**VALID: Decision-Support Platform (CUCAI Responsible AI Award)** | *Python, FastAPI, BERT, XGBoost*

- Led a **10-person** team to ship a multimodal decision-support backend, winning CUCAI 2026's Responsible AI Award with **91% precision** and **88% macro-recall** on **120k+** unstructured records.
- Designed a **FastAPI** gateway around a BERT labeling pipeline and an ensemble of XGBoost, MLP, and Random Forest models, exposing versioned REST endpoints for downstream clients.

**Disaster Detection App (MEC 2nd Place)** | *Swift, iOS, REST APIs, Twilio*

- Built an iOS app that fuses on-device sensors with a disaster-dataset API to detect hazardous events and trigger emergency alerts in **under 5s**, using Twilio SMS and local caching for reliable operation during connectivity loss.

**KareBear (ElleHacks 2nd Place / 100+ teams)** | *React Native, TypeScript, GCP*

- Shipped a cross-platform Expo app with **GCP** integrations for real-time audio and visual processing, plus a conversational backend, placing **2nd of 100+** teams while delivering the full application under hackathon time constraints.

## Technical Skills

**Languages**: Python, C/C++, Swift, Java, JavaScript, TypeScript, SQL, Bash, MATLAB
**Software & Systems**: REST APIs, FastAPI, Flask, React Native, Node.js, Spring Boot, Linux, eBPF, Pytest, JUnit
**Infrastructure**: Kubernetes, OpenShift, Docker, Podman, Helm, Ansible, Jenkins, GitLab CI/CD, Git
**Cloud & Data**: AWS (S3, EC2), GCP, PostgreSQL, Supabase, Firebase
**ML (supporting)**: PyTorch, TensorFlow, Hugging Face, Scikit-learn